

EES 102: Introduction to Environmental Sciences

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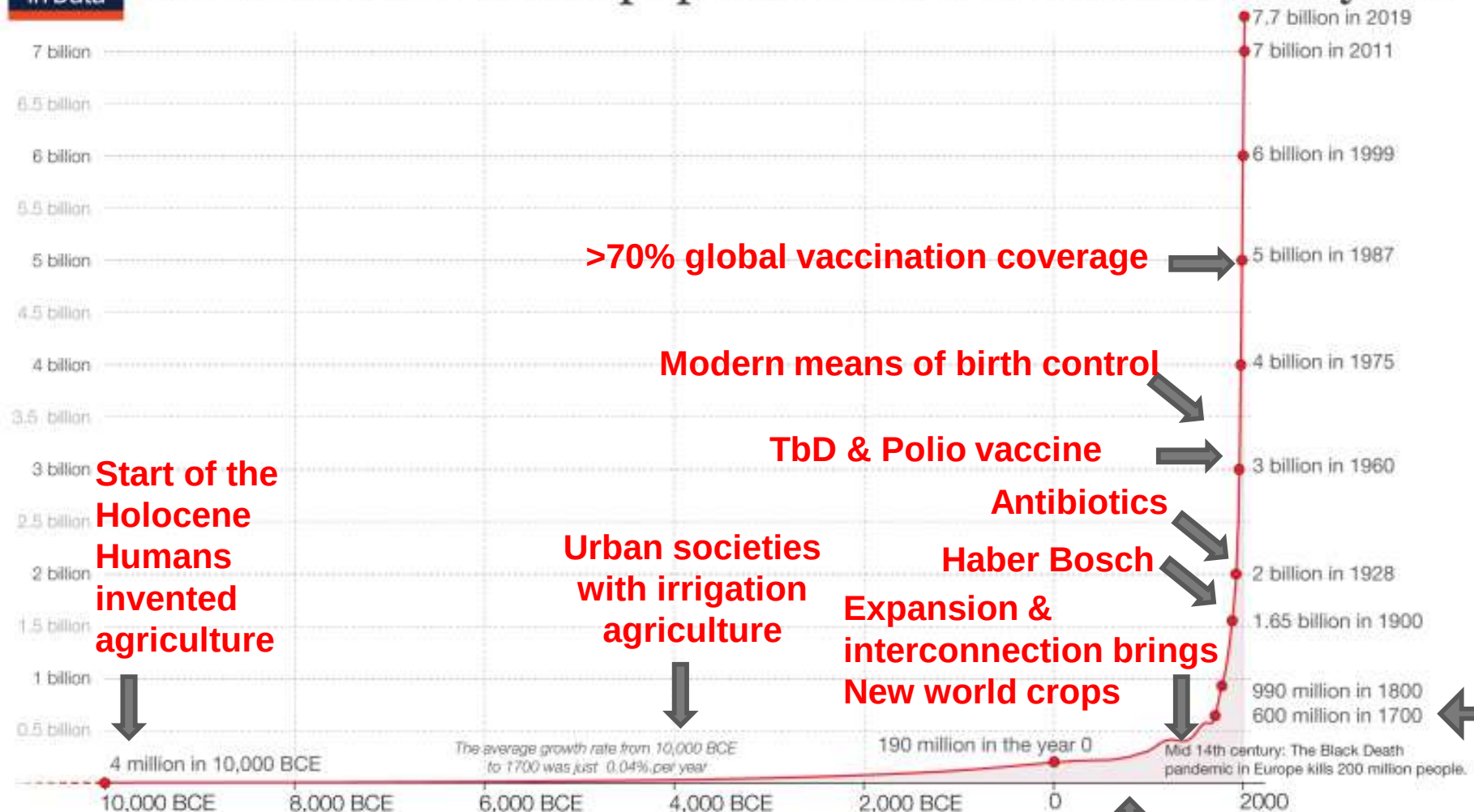


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THE ANTHROPOCENE

HOW IT STARTED....

Our World in Data The size of the world population over the last 12,000 years



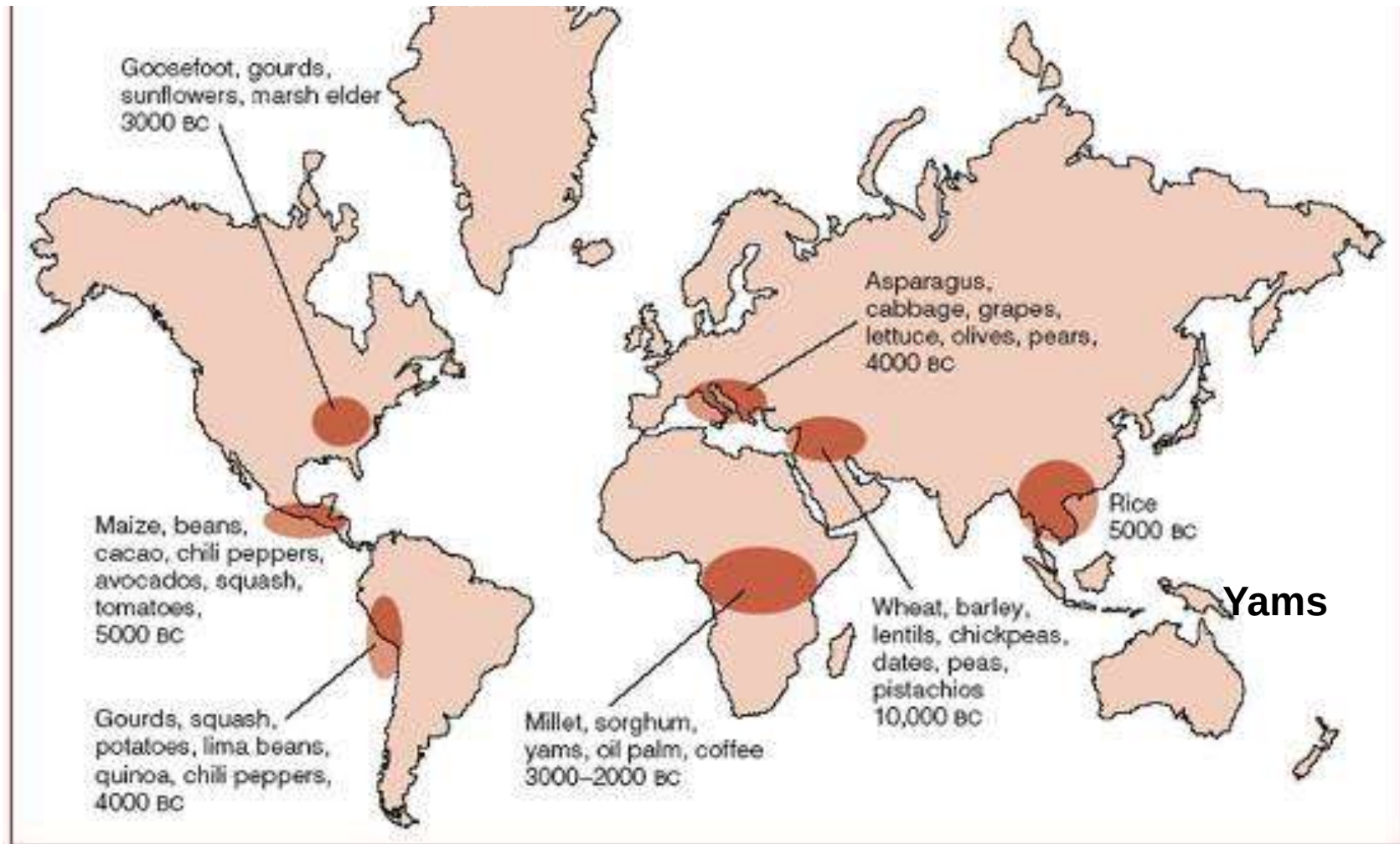
Based on estimates by the History Database of the Global Environment (HYDE) and the United Nations. On OurWorldinData.org you can download the annual data. This is a visualization from OurWorldinData.org, where you find data and research on how the world is changing. Licensed under CC-BY-SA by the author Max Roser.

First formulation of germ theory and introduction of smallpox vaccination in middle East and India allows larger cities

Germ theory & smallpox vaccination Europe

EMERGENCE OF AGRICULTURE

All the energy that supports the biosphere comes from sunlight and is used by plants in a process called photosynthesis. Farmers are diverting more of that energy into species that they can use and away from all of the other species. Agriculture is in the end a huge energy grab by one species: Humans



EMERGENCE OF AGRICULTURE

- **Why did Agriculture emerge in so many places at roughly the same time? Two factors contributed**
- **1) Overpopulation:** During the Palaeolithic when populations grew too much homo sapiens sapiens migrated into new areas, eradicating homo neanderthalensis, denovisians and other hominide species. After the settlement of the Americas ~15,000 years ago, there was no more place left to go, so humans had to learn how to get more resources from a given area
- **2) Climate change:** During the Paleolithic (Ice age) climate was cold and unpredictable. When the world entered the Holocene about 12,000 years ago, the climate became warmer and wetter and more stable. Farming became both possible and necessary. The icy steppes disappeared and the large game that humans had specialized in hunting went extinct. At the same time, huge areas were flooded by rising sea levels while other areas saw an increase in plant and animal biodiversity which allowed humans to stay in one and the same place year around. However, sedentary lifestyles accelerated population growth. To provide resources to a growing population humans started to water and select plants, clear weeds and breed animals. They started to farm.

ACCELERATION OF COLLECTIVE LEARNING

- **When agriculture appeared, history seemed to speed up. Five thousand years after the appearance of agriculture, the number of humans had increased almost 10-fold, from just a few million to almost 50 million.**
- This rapid growth was made possible by an increase in the number of innovations. Farmers spread into wooded and semi-arid zones, designed new types of buildings, discovered ways to domesticate animals, pioneered new uses for clay and metals, and began to develop simple forms of irrigation.
- With the appearance of agrarian civilizations, the rate of innovation increased. Architects designed monumental architecture like pyramids and temples; metalworkers made refined tools and weapons; writing appeared, providing more reliable ways of storing and preserving information.
- With the domestication of horses and camels, goods and people were moved over large distances; meanwhile, sailors and merchants figured out how to travel across the Indian Ocean and into the Pacific Ocean. There was also innovation in government and lawmaking, in the types of religion, and in art and literature. New goods, including glassware, jewelry, and the coins that enabled trade and taxation to evolve, were made.

NETWORK SIZE AND DIVERSITY MATTERS FOR LEARNING

- **Collective learning was accelerating by agriculture.** To understand how and why, we need to think about how humans exchange ideas. When several individuals are linked, they form a “network.” Networks appear in many different varieties. The Internet is a network of computers; economies are networks of individuals who are buying and selling; proteins within a cell form networks linked by different chemical reactions; the electricity grid is a network. All networks contain two main kinds of things: points, and links between the points.
- **The First Rule About Networks: Size Matters!** How many possible links are there between three friends? The answer is three. Between four friends? The answer is not four; it is six. You can see this by drawing lines between four points. How many links are there between five friends? The answer is 10. We can calculate the relationship between nodes and edges precisely using a mathematical formula but the main trend is that, as groups get larger, the number of possible links within the network increases *much* faster than does the number of people. So in large groups, the possibilities for sharing information are much greater than in small groups. In the Paleolithic era, most communities were tiny, usually with fewer than 40 or 50 people. Much larger communities — villages with, say, 2,000 inhabitants — developed with the appearance of agriculture. A village this size would produce a network of 2 million possible links.
- **The Second Rule About Networks: Diversity Is As Important As Size!** If everyone lives and thinks exactly like everyone else, there won't be much new information to exchange. When different groups that have slightly different ways of providing food and shelter, slightly different forms of clothing, different rituals, and different stories and they get connected through trade, collective learning accelerates. Larger communities can also produce more consistent agricultural surplus which encourages division of labour or specialization.

LARGE NETWORKS ARE INHERENTLY UNEVEN

- **As networks get larger, the amount of connectedness of any single node compared with another node gets more and more uneven.**
- We can see the same pattern in agrarian civilizations: isolated villages had limited connections; townships with markets or temples were much better connected; and capital cities were connected to the entire kingdom.
- **The uneven distribution of information and connectedness can help us understand why wealth and power are distributed so unevenly in agrarian civilizations.** An individual connected to 10 other individuals can form alliances or teams with those individuals. But an individual connected to 1,000 other individuals has access to more information and can form greater and more powerful alliances. (This is perhaps why we say of powerful people that they are “well connected.”) More connectedness seems to mean not only more information but also more wealth and power.
- **It is not by random chance that the wealthy and powerful individuals in society maintain control on the storage and dissemination of information.**

IRRIGATION AGRICULTURE

- With increasing population pressure farmers started to farm in steppe soils which were incredible fertile. However farming there was only possible with the help of irrigation. This required construction of sophisticated irrigation networks. Agricultural societies became more complex. A system of power brokers, elites which organized the construction and maintenance of these irrigation systems appeared.

Farmers (90%)
extract resources
from the
environment



Elites extract
resources from
farmer. If necessary
with the use of force

Hierarchies
Monumental architecture
Armies (& paid enforcers)
A tax system
Script

Urbanization intensified environmental problems
Such as contaminated drinking water
& irrigation cause soil salinity issues

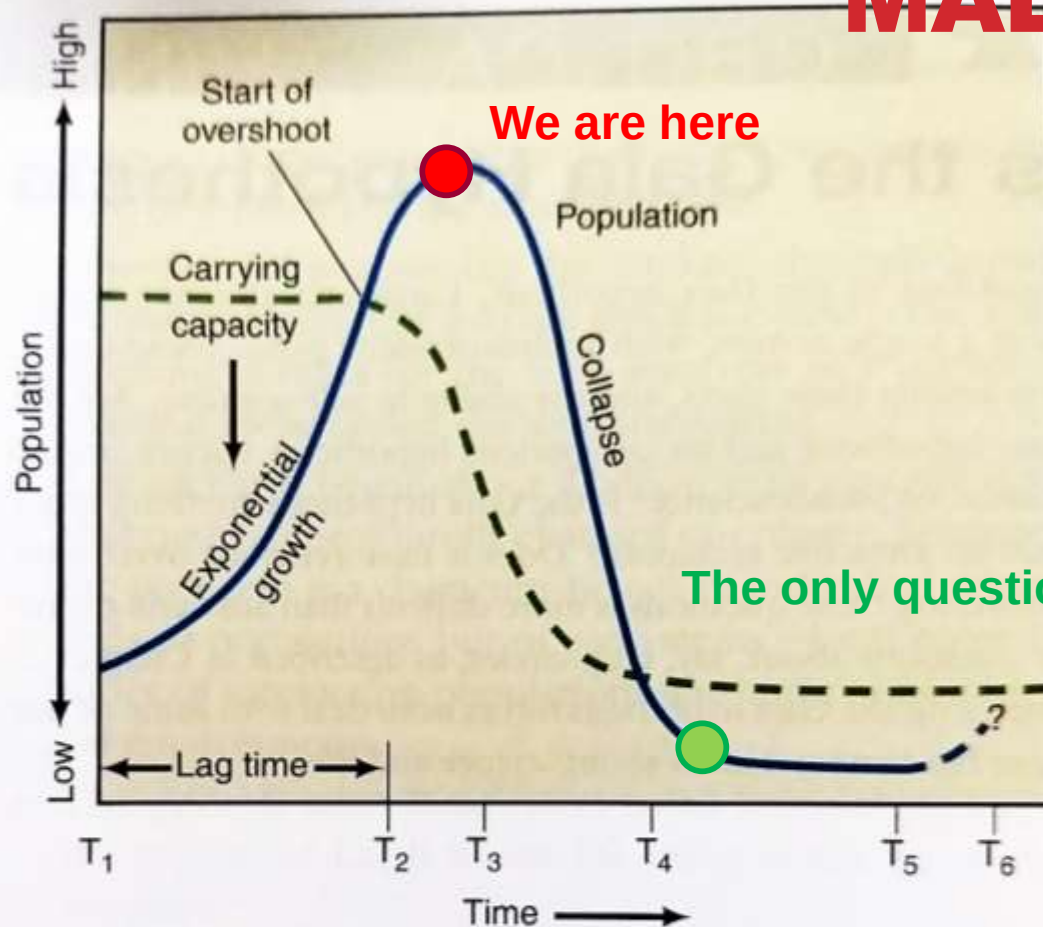
MIGRATIONS AND INTENSIFICATION

- If we want to prevent our bodily complexity and all the complexity that we have created into chaos, we must keep harvesting matter and energy flows on a regular basis.
- In most of human history the quest for sufficient matter and energy to survive and reproduce has been the overarching theme.
- Innovations such as the domestication of cattle, goats, sheep and camels, the use of milk or the four crop rotation in early modern Europe and the Haber Bosch process slowly raised the carrying capacity of earth for sustaining a growing population. However, throughout much of the human history, innovation did not keep pace with population growth.
- Just like any other animal in nature we used to breed till we strained the resources to the point where the population could no longer be fed and supported with the technology of the time.
- Every couple of generations (typically every 200-300 years), population in a once prosperous area would reach unsustainable levels resulting in periods of infighting (revolutions or civil wars), wars between countries, famines and epidemics.
- These periods would bring a population decline and after some time populations would recoil.

MALTHUSIANISM

- Malthusianism is the theory that population growth is potentially exponential, according to the Malthusian growth model, while the growth of the food supply or other resources is linear, which eventually reduces living standards to the point of triggering a population decline.
- This event, called a Malthusian catastrophe occurs when population growth outpaces agricultural production, causing famine or war, resulting in poverty and depopulation.
- These concepts derive from the political and economic thought of the Reverend Thomas Robert Malthus, as laid out in his 1798 writings, *An Essay on the Principle of Population*. Some economists contend that since the industrial revolution in the early 19th century, mankind has broken out of the trap. Others argue that the continuation of extreme poverty indicates that the Malthusian trap continues to operate

MALTHUSIANISM



The only question is how we get to down here

Figure 3.14 ■ The concept of overshoot, illustrating the influence of exponential growth, lag time, and collapse on carrying capacity. Carrying capacity starts out relatively high, but as exponential growth increases the population beyond the carrying capacity, overshoot occurs and population collapses. If environmental damage occurs as a result of overuse and damage to resources on which the carrying capacity depends, then the carrying capacity also crashes, as shown here. (Source: Modified after D.H. Meadows and others 1992.)

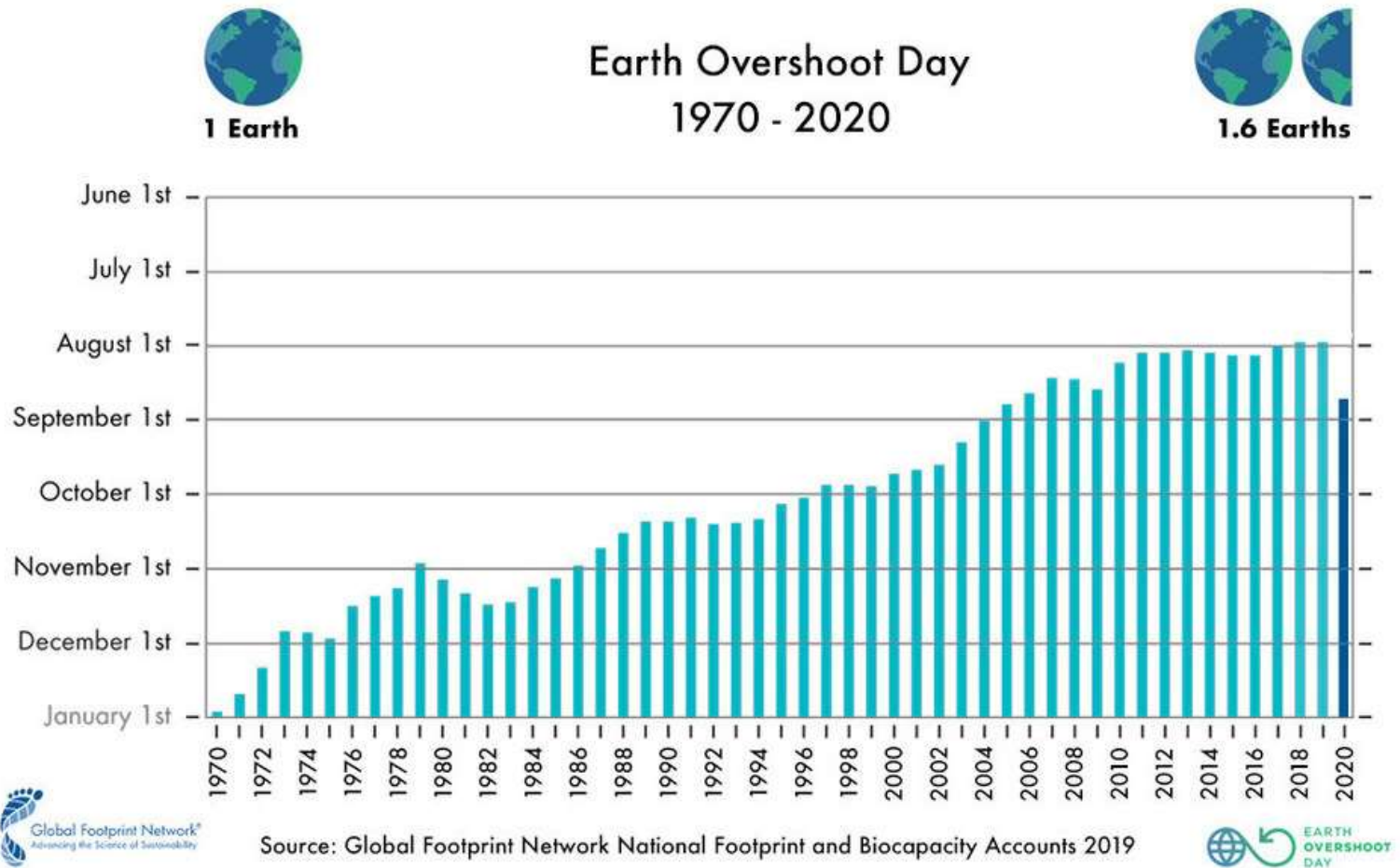
EARTH OVERSHOOT DAY OR WHY ENVIRONMENTAL PROTECTION IS URGENT

- **Earth Overshoot Day (EOD) is the calculated illustrative calendar date on which humanity's resource consumption for the year exceeds Earth's capacity to regenerate those resources that year.** The term "overshoot" represents the level by which human population overshoots the sustainable amount of resources on Earth. When viewed through an economic perspective, EOD represents the day in which humanity enters environmental deficit spending. EOD is calculated by dividing the world biocapacity (the amount of natural resources generated by Earth that year), by the world ecological footprint (humanity's consumption of Earth's natural resources for that year), and multiplying by 365, the number of days in a year:

$$\text{world biocapacity} / \text{world ecological footprint} \times 365 = \text{EOD}$$

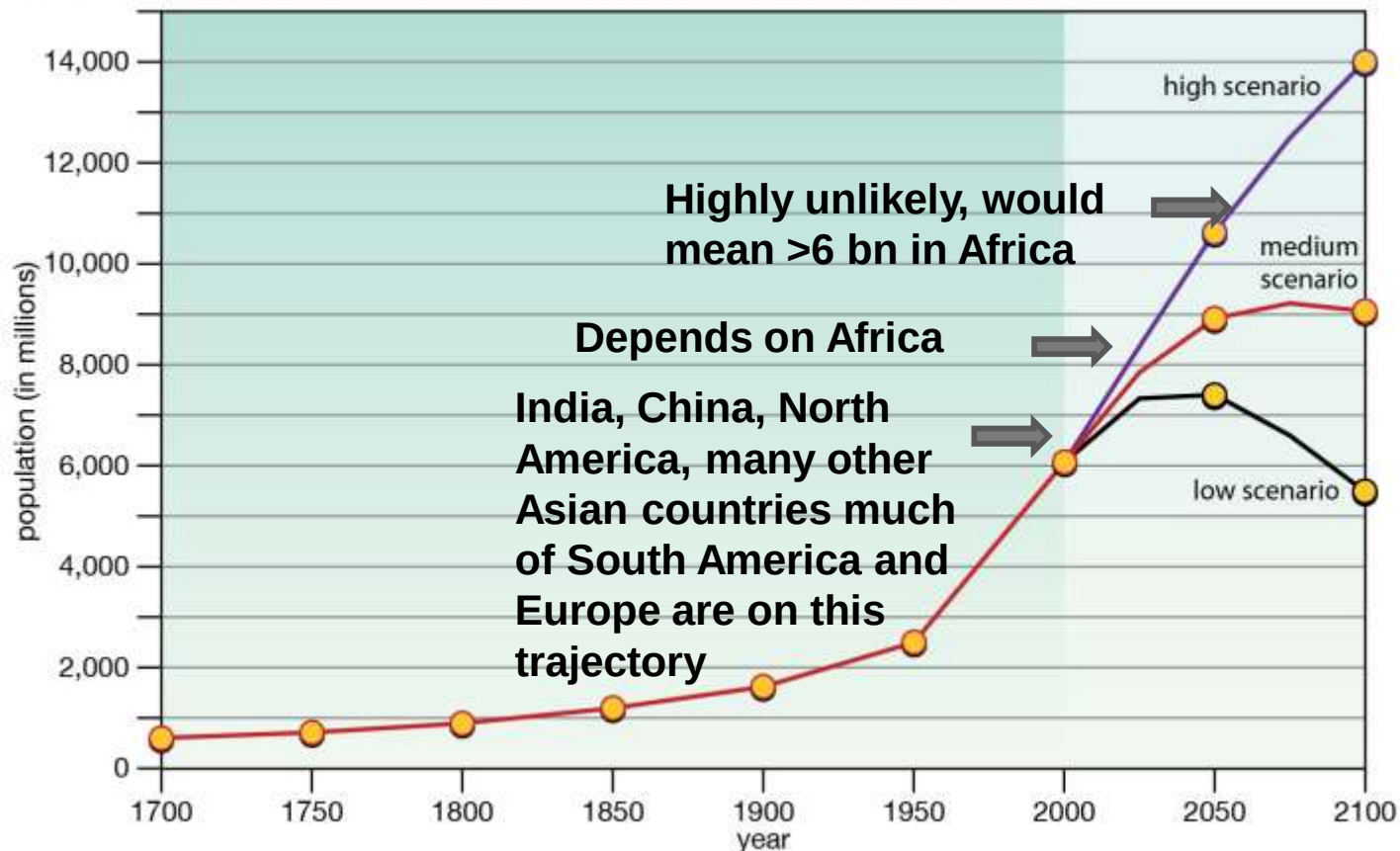
- In 2023 EOD landed on August 2
- **OVERSHOOTING OF THE ECOLOGICAL CAPACITY PRECEDES THE MALTHUSIAN POPULATION CATASTROPHE**

EARTH OVERSHOOT DAY



There is a more pleasant way get to sustainable population sizes

World population (1700–2000) and population projections (2000–2100)

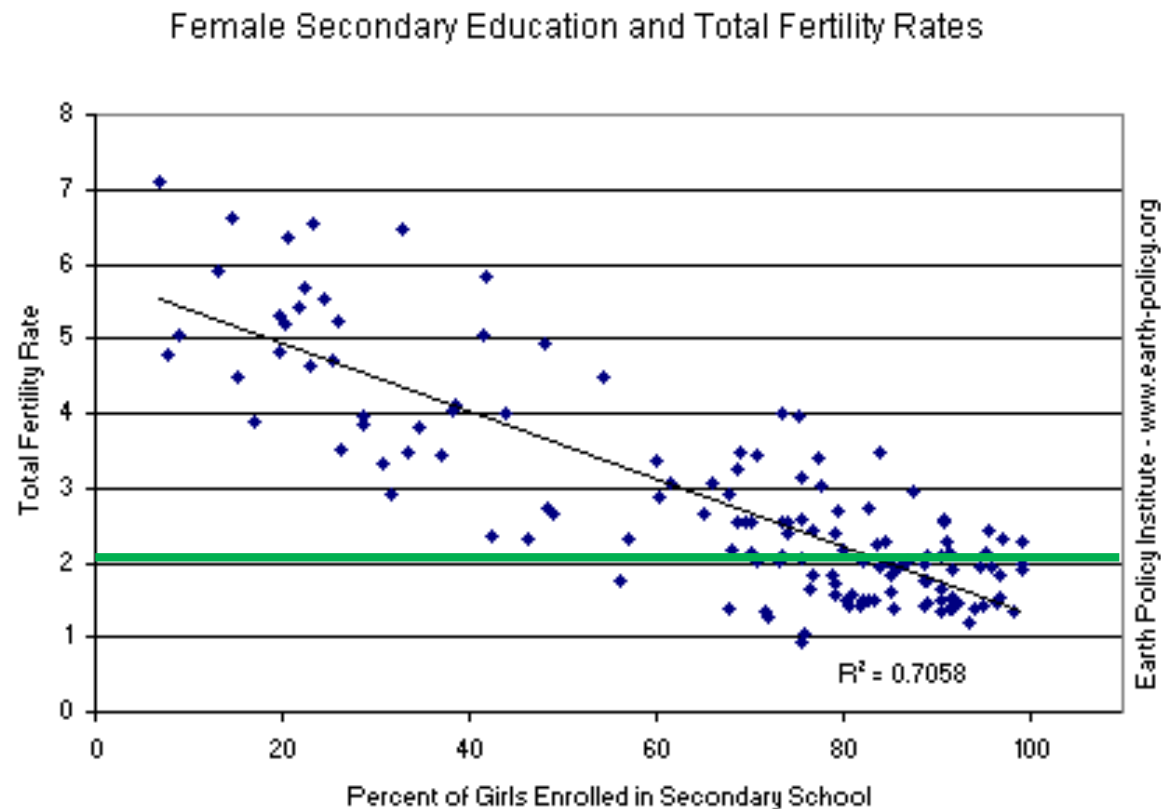


Source: United Nations Department of Economic and Social Affairs/Population Division 2004

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Modern means of birth control allow us to choose the number of children we want. For the first time we can reduce our population to sustainable levels without riots, wars and a global pandemic.

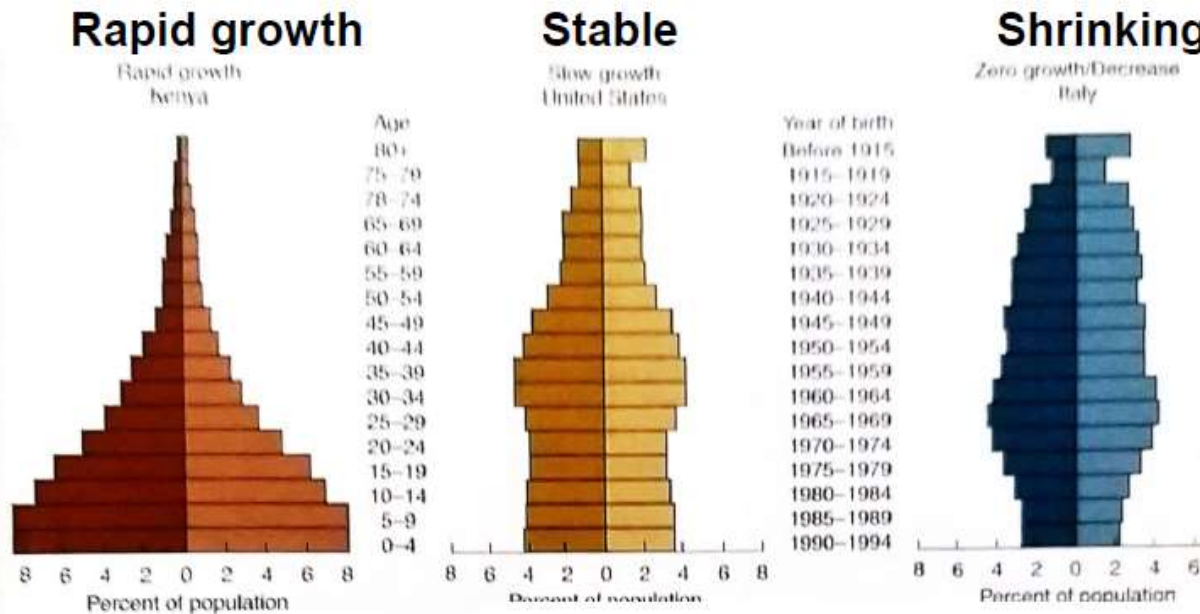
Female education leads to low fertility



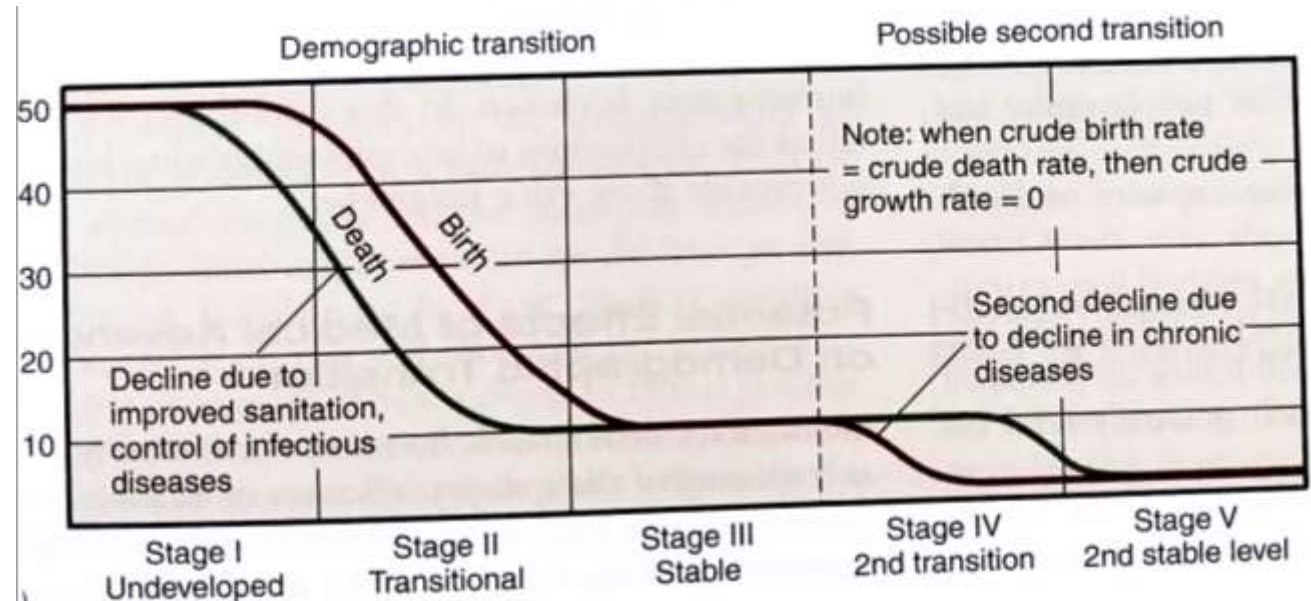
Source: EPI from UNESCO

Getting girls to school and retaining them till the secondary level (post puberty) enables them to assert their reproductive rights and generally results into stable or shrinking populations.

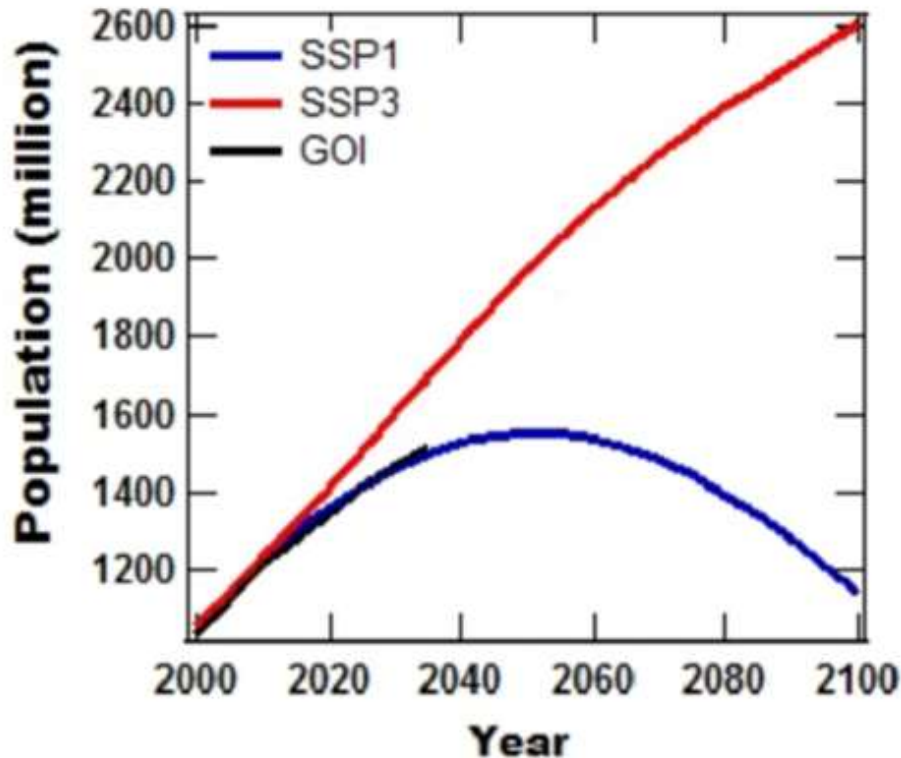
THE DEMOGRAPHIC TRANSITION



birth control is adopted only when parents are sure that their children are going to survive



THE RTE ACT INTRODUCED IN 2006 HAS STARTED PAYING OFF



India is reaping the benefit of increased school retention of girls into middle school

We will have to find a sustainable way to provide a better life for about 1.5 billion people ...

But not for 2.6 billion and that is great news!

THE BIG ENERGY STEPS



Renewable energy comprises of a number of techniques to capture the energy of the sun more directly without going through the process of photosynthesis. Photosynthesis captures only 1-2% of the incident energy so it is easy to see that a solar panel with 20-40% efficiency requires much less space. Novel materials can also help us to dispose excess heat ($>100 \text{ W/m}^2$) directly to space through the atmospheric window. This allows us to get more energy from less land and consume less for cooling. This can lead us into an era where the equation: increased energy consumption = increasing destruction of natural ecosystems

need not hold any longer. **More efficient & less destructive energy harvest can get us out of the Anthropocene. Anyway fossil fuels are finite - no alternatives!**